

! Memory management terms

! Frame:

A fixed-length block of main memory. Main memory is divided into these equal-sized blocks.

Page:

A fixed-length block of data that is stored in secondary memory (e.g. disk). Pages are copied into frames when needed.

Segment:

A segment is a larger, variable-sized block of data in secondary memory. It can be copied into the main memory as a whole or divided into pages and copied one by one.

Memory management requirements

Memory management is designed to satisfy these requirements:

→ Relocation → Protection → Sharing → Logical organization → Physical organization

! Relocation:

Programs don't know where they will be in memory when running. To use the processor well, processes are moved in and out of memory. Sometimes they need to be relocated to different places.

Protection:

Processes need permission to read or write memory locations. The position of a program in main memory cannot be predicted. Memory access must be checked while the program is running.

Sharing: Advantageous to allow each process access to same copy of program rather than have their own separate copy. The systems that allow relocation also support sharing.

Logical Organization: Memory is organized as linear.

- Programs are written in modules

Physical Organizations: Programmers can't manage memory directly.

Programmers are unaware of the exact memory space available.

- **Overlaying:** Allows multiple modules to share the same memory region but, time consuming to implement.

Memory Partitioning (Memory Ayırması)

Memory management brings processes into main memory so the processor can execute them. This involves virtual memory and uses methods like segmentation and paging.

Memory Management Techniques



Fixed Partitioning

The main memory is divided into fixed-size parts when the system starts. Each process is loaded into a partition of equal or larger size.

- **Strength:** Easy to implement, low system overhead.
- **Weakness:** Wastes memory due to internal fragmentation, and limits the number of active processes.



Dynamic Partitioning

Partitions are created dynamically, matching the size of each process.

- **Strength:** No internal fragmentation and better memory usage.
- **Weakness:** Processor usage is inefficient due to the need for compaction to handle external fragmentation.

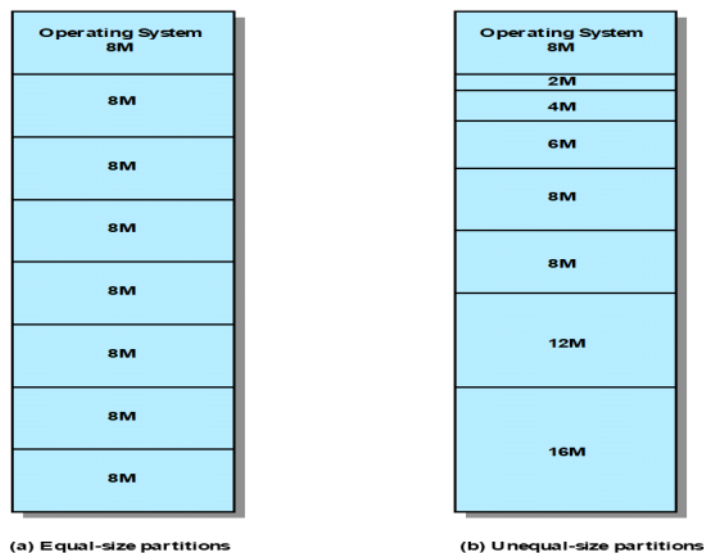


Figure 7.2 Example of Fixed Partitioning of a 64-M byte Memory

Disadvantages of equal-size fixed partitions (Sabit Bölümleme)

- A program might be too large, needing overlays.
- Memory is not efficient because small programs still take a full partition.
- **Internal fragmentation** happens when there is wasted space because the program is smaller than the partition.

Placement Algorithm

Best-Fit : Select the block closest in size to the process.

First-Fit : Starts from beginning of memory and picks the first block that is larger enough.

Next-Fit : Starts searching from the last used position and picks the next block that is larger enough.

Dynamic Partitioning

In dynamic partitioning, partitions have different sizes and are created as needed. A process is given exactly the amount of memory it requires, which improves memory usage.

Example & Problem: At first, memory is used efficiently by giving processes just enough space. However, over time, small empty spaces (holes) appear in memory, which cannot be used by new processes. This problem is called **external fragmentation**, meaning memory becomes divided into many small unused parts, reducing overall efficiency.

- **External Fragmentation(dışsal parçalanma):**

- Memory becomes divided into small unused parts.
- This reduces overall memory efficiency.

- **Compaction(birleştirme):**

- A method to fix external fragmentation.
- The OS moves processes to make free memory into one large block.
- This process takes a lot of time and uses CPU heavily.

Addresses

- **Logical Address:**

A memory reference that does not depend on the actual memory location.

- **Relative Address:**

A type of logical address that shows a location relative to a known point.

- **Physical or Absolute Address:**

The real location in main memory.

Paging:

- Paging divides memory into equal fixed-size blocks called **frames**.
- A process is also divided into equal-sized parts called **pages**, which are loaded into available frames.
- This method allows processes to be stored in non-contiguous memory locations, helping to avoid external fragmentation.

Page Table:

- Managed by the operating system for each process.
- Keeps track of which frame contains each page of a process.

Segmentation (B1 Summary):

- A program is divided into **segments** of different lengths.
- Each address has two parts: **segment number** and **offset**.
- Similar to dynamic partitioning but avoids internal fragmentation.
- Helps organize programs and data, making it easier for the programmer.
- Limitation: The programmer must know the **maximum segment size**.